

MARCELO VINICIUS TAPARELLI

AI Engineer & Software Engineer

Python · RAG · Agents · Evals · TypeScript · Bun · Node.js · PostgreSQL · AWS

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PROFESSIONAL SUMMARY

Product-minded AI Engineer & Software Engineer building Generative AI and backend systems with RAG, agents and evals. Typed contracts, tests and observability support decisions in security, reliability, cost, latency and operability.

PROFESSIONAL EXPERIENCE

Agência Catus | Full Stack Developer | Jul 2026 - Present

- Develop and evolve the agency's new website and two WordPress blogs, including Custom Post Types, theme customization, and new features.
- Maintain an e-commerce implementation on the Wake platform, troubleshoot production issues, and work on performance/Core Web Vitals.
- Deploy directly to server environments and continuously improve web applications.

EVAG Comunicação e Tecnologia | Developer | Apr 2024 - Present

- Developed Google Drive-to-WordPress automation for generating petition pages, reducing manual steps and making the publishing workflow more direct.
- Developed an internal request management solution integrated with GitLab, supporting organization, traceability, and flow of operational demands.
- Develop and maintain systems, websites, and automation with Laravel, PHP, Python, JavaScript, WordPress, and Tailwind, including production fixes and new features.

EVAG Comunicação e Tecnologia | Product Owner | Apr 2023 - Mar 2024

- Managed website delivery from discovery through delivery, connecting clients, design, and development.
- Structured requirements and support workflows, prioritizing across business needs, user experience, and the team's technical capacity.

TECHNICAL SKILLS

AI Engineering / Artificial Intelligence: Python, FastAPI, LLMs, RAG, embeddings, pgvector, hybrid retrieval, LangGraph, OpenAI, agents, evals, structured outputs, human-in-the-loop, guardrails, OpenTelemetry, PyTorch, CUDA and fine-tuning/domain adaptation.

Software Engineering: TypeScript, Bun, Node.js, JavaScript, PHP, Laravel, Express, Prisma, REST APIs, PostgreSQL, Redis, Docker, Terraform, Git, CI/CD with GitHub Actions; AWS lab with ECS/Fargate, ALB, RDS, ElastiCache, ECR, IAM, Secrets Manager and CloudWatch.

Engineering practices: Product thinking, Clean Architecture, DDD principles, SOLID, Clean Code, TDD, microservices fundamentals, unit/integration/E2E tests, security, reliability, observability and performance.

SELECTED PUBLIC PROJECTS

OpsPilot AI: Python/FastAPI, evidence-grounded RAG, pgvector/hybrid retrieval, OpenAI and LangGraph agentic AI. Deterministic policies, hash-bound human approval, GitLab reconciliation, evals and OpenTelemetry. Independent post-release OpenAI and GitLab sandbox live smokes; validated Terraform blueprint, without AWS deployment.

Ops Triage AI: Deterministic baseline, local LLM, hybrid policy and human review. Jev/Laya evaluation-only on a frozen synthetic held-out (70 tickets). Laya with PyTorch/CUDA: exact tuple 11.43% zero-shot → 85.71% adapted; validation-based selection. HIGH/CRITICAL recall regressed from 92.86% to 85.71%.

Resilient Transaction API: TypeScript/Bun, PostgreSQL, Redis, concurrent idempotency, rate limiting, retries, circuit breaker, observability and Docker. Terraform and a temporary non-production AWS lab, with functional validation, destroy and resource audit.

Salus: Node.js/Express, PostgreSQL/Prisma, Clean Architecture, domain modeling and a manually implemented TDD core; JWT authentication, ownership-based authorization and automated tests.

EDUCATION

Postgraduate Program in AI Engineering — Cruzeiro do Sul Virtual

In progress · expected completion: Apr 2027

Systems Analysis and Development — UniBF

Completed in 2026

Software Engineering — Alura

Back-end · Node.js APIs · Authentication · Testing · Security · DevOps/CI/CD · Cloud/AWS · Secure Development

LANGUAGES

Portuguese — native · English — advanced